

Introduction & Hook

“If a car moves in VR, should its wheels move automatically?”

That automatic behavior happens because of **hierarchy in scene graphs**.

Today we study how VR systems organize objects geometrically.

PART 1: Geometry in VR

Geometry Basics

In VR, geometry represents the **shape of objects**.

Objects are made of:

- Vertices
- Edges
- Faces (triangles)
- Meshes

Example:

- A cube = 8 vertices + 12 triangles
- A sphere = many small triangles

Explain:

All VR objects are built from geometric primitives.

PART 2: Scene Graph

What is a Scene Graph?

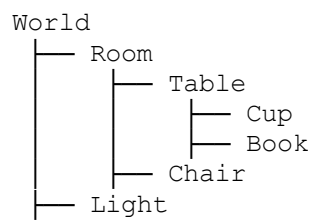
Definition:

A scene graph is a hierarchical tree structure used to organize objects in a virtual scene.

It represents:

- Objects
- Transformations
- Relationships
- Rendering order

Draw on board:



Explain:

- World is root node
 - Objects are nodes
 - Each node can have children
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Why Scene Graph is Important

- Efficient rendering
 - Easy transformation management
 - Hierarchical movement
 - Organized scene structure
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PART 3: Parent–Child Relationships



30–45 min → Parent–Child Concept

Definition:

In a scene graph, a parent object controls the transformation of its child objects.

Example:

Car (Parent)

- Wheel (Child)
- Door (Child)
- Headlight (Child)

If car moves:

→ Wheels move

→ Door moves

→ Headlights move

This is called hierarchical transformation.

Key Concept: Inherited Transformation

If parent rotates:

→ Child rotates automatically.

If parent scales:

→ Child scales automatically.

But child movement does not affect parent.

Example in VR:

VR Hand Model:

Hand (Parent)

- Fingers (Children)
- Thumb (Child)

If hand moves:

→ Fingers move automatically.

If finger moves:

→ Hand does NOT move.

Transformation Propagation

Explain transformation flow:

Local Space $\rightarrow$ Parent Transform $\rightarrow$ World Transform

Each child's world position is calculated as:

Parent transformation $\times$ Child local transformation

This reduces computation complexity.

Advantages of Scene Graph Hierarchy

- Logical organization
 - Easy animation
 - Efficient rendering
 - Simplified collision detection
 - Better scene management
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Recap + Questions

Recap:

- Geometry defines shape
 - Scene graph organizes objects hierarchically
 - Parent controls child transformations
 - Transformations propagate downward
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Quick Oral Questions:

- What is a scene graph?
- What happens if parent object rotates?
- Can child affect parent transformation?
- Why is hierarchy important in VR?

